Block C
Group 21

Error Analysis and Iterative Improvements of DeBERTa

The best-performing model in this project was built on the **DeBERTa v3 XSmall** architecture, fine-tuned in a multi-task setup using the ***microsoft/deberta-v3-xsmall*** checkpoint**,** to classify emotions from television show transcript. It predicts three emotion-related tasks simultaneously: **primary emotions** (e.g., *Happiness*, *Anger*), **secondary emotions** (e.g., *Pride*, *Curiosity*), and **emotional intensity** (*Mild* to *Overwhelming*). Chosen for its strong contextual understanding and efficiency, this model achieved the highest F1 score on the test set for primary emotion classification and showed strong performance in capturing both broad and nuanced emotional signals.

This document presents a focused **error analysis**, beginning with a brief look at dataset quality and initial model behaviour. We then assess the model's performance across each iteration and summarize key error patterns. These findings highlight how our iterative changes improved predictions and reveal remaining challenges, informing future improvements and practical use.

Dataset Quality Assessment

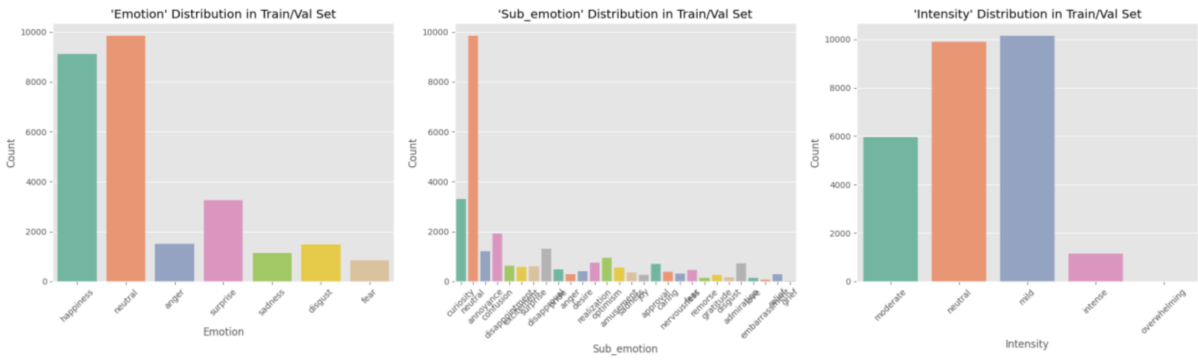

Figure 1: Train Dataset Distribution

The model was trained on a merged dataset from all project groups and evaluated on our group's original test set. Across all tasks — *emotion*, *sub_emotion*, and *intensity* — the dataset shows strong class imbalance, as you can see in ***Figure 1*** above, with ***neutral*** and ***happiness*** making up nearly **70%** of the training data, and ***fear***, ***sadness***, and ***disgust*** together under **13%**. The imbalance ratio between majority and minority classes reaches 11.5:1. To address this, we applied class weighting, giving higher importance to underrepresented classes (e.g. *surprise*: 4.54, *fear*: 1.19).

Further analysis revealed structural and linguistic biases. Minority emotions had greater vocabulary diversity (up to **42%**), while majority classes used more repetitive language. ***Neutral*** texts were shorter, and sentiment scores aligned with emotional polarity (***happiness*: +0.15; *sadness*, *anger*, *disgust*: ~–0.04**). ***Figure 2*** shows that sub-emotion and intensity labels were unevenly distributed, with ***neutral*** having only one sub-emotion and all samples labelled as neutral intensity. Vocabulary overlap between emotions was low (**15–35%**), highlighting distinct expression styles. These limitations pose key challenges for learning emotional nuance and generalizing across emotion categories.

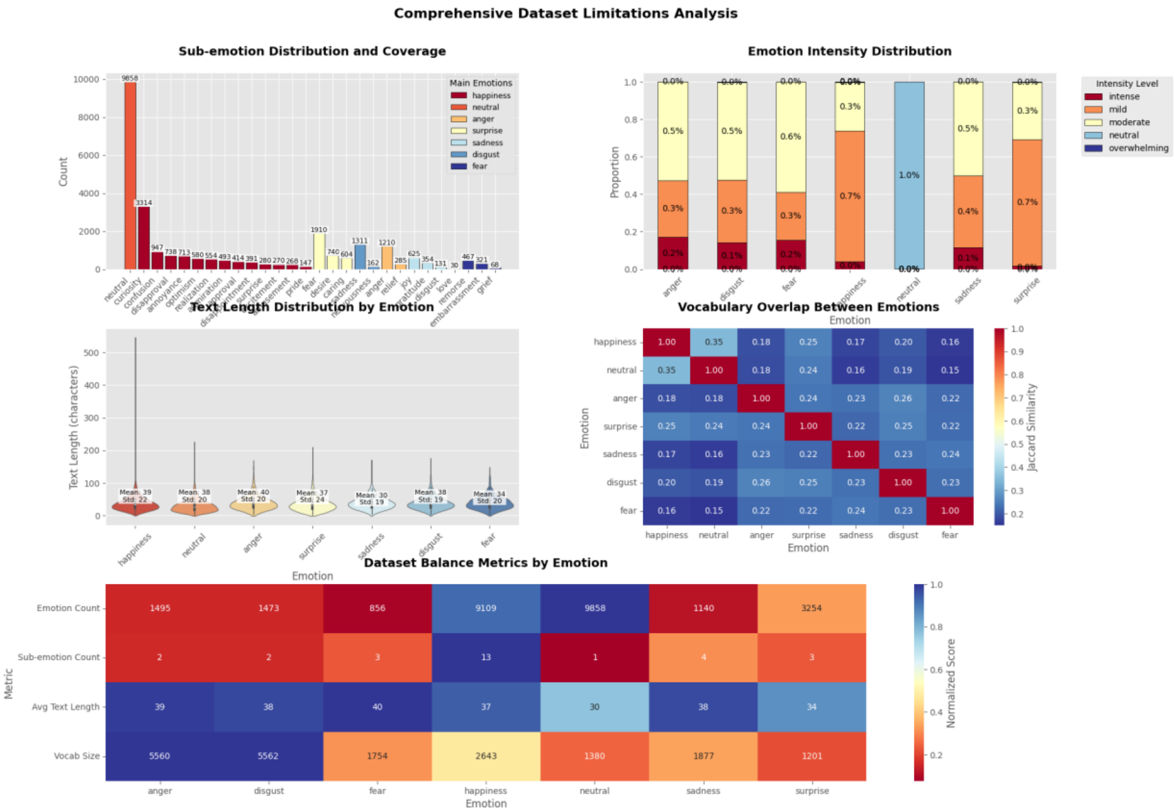

Figure 2: Dataset Limitations Analysis

Iteration 1

In Iteration 1, we established a baseline by training the model on **raw text** only, without any feature extraction inputs. The goal was to map **"text"** to **"emotion", "sub-emotion",** and **"intensity"** while using **class weighting** to reduce bias toward dominant labels.

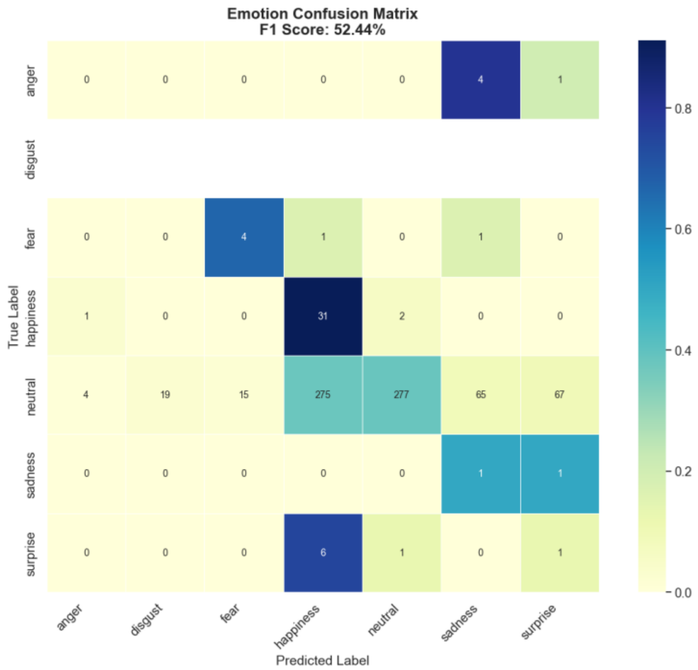

Figure 3: Emotion Confusion Matrix - Iteration 1

The baseline model performed best on sub-emotion prediction (**F1: 86.6%**), with moderate results for emotion (**52.4%**) and intensity (**41.5%**). ***Figure 4*** below shows that accuracy was highest for **happiness (91%)** and **fear (67%)**, but extremely low for **surprise (12%)** and **anger (0%)**. In the intensity task, **neutral** was predicted most reliably (**78%**), while **intense** cases saw **0%** recall. Combined task accuracy dropped to **22%**.

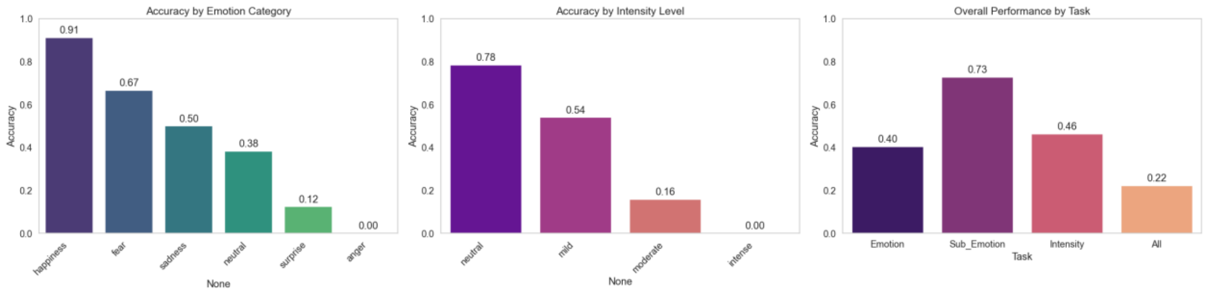

Figure 4: Accuracy - Iteration 1

The emotion confusion matrix in ***Figure 3*** reveals major confusion between **neutral**, **happiness**, and **sadness**, with neutral often misclassified as **happiness**, **fear**, or even **disgust**, despite correct sub-emotion predictions. Sub-emotion outputs heavily defaulted to **neutral**, while nuanced categories such as **pride** or **nervousness** were frequently missed. Intensity predictions also skewed toward **mild** and **neutral**, with **moderate** and **intense** cases commonly misclassified. This indicates that the model struggles to capture **emotional nuance and gradation**, particularly in underrepresented or **subtly expressed** cases—for example, the neutral sentence *"Let me hear you say, I want to be here..."* was misclassified as **happiness**, reflecting difficulty in distinguishing emotionally ambiguous content.

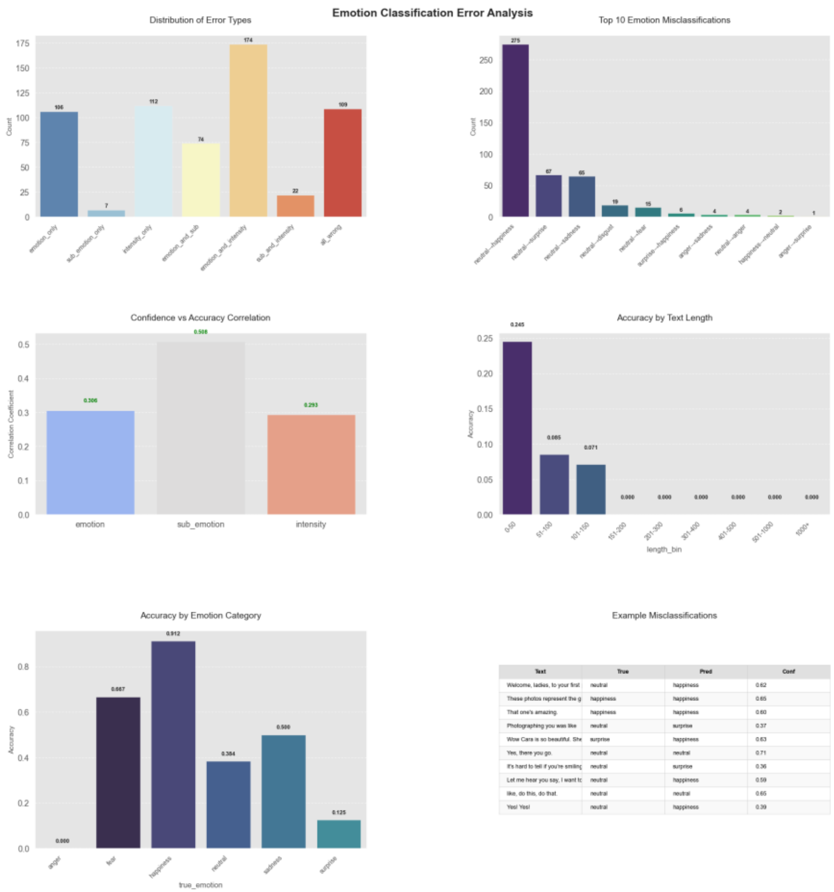

Figure 5: Error Analysis – Iteration 1

Further analysis from ***Figure 5*** above confirmed that **neutral-to-happiness confusion** was the most frequent error, with **275 instances** and an **average confidence** of **0.552**, often triggered by descriptive or mildly positive language. Overall, **73.7% of all errors** involved the **neutral** class, marking it as the most problematic. The model often predicted **"happiness"** for neutral cases with positive tone, even when **sub-emotion and intensity** were correct. Accuracy was highest for **very short texts** (under 50 characters) but **dropped to zero** beyond 150 characters, showing sensitivity to input length. **Confidence-to-accuracy correlation** was strongest for **sub-emotion** (0.508), with lower values for **emotion** (0.306) and **intensity** (0.293).

In terms of **hierarchical performance,** demonstrated by ***Figure 4***, **sub-emotion** predictions were the most accurate (**0.727**), while **intensity** lagged behind (**0.463**), with **intense cases entirely missed**. Predictions skewed toward **mild and neutral**, indicating low sensitivity to strong emotions. These patterns highlight the need for **data augmentation**, **contrastive learning**, and **hierarchical attention** to better handle nuanced or overlapping emotional expressions.

Iteration 2

In the second iteration, we extended the baseline by integrating **linguistic and semantic features**—including **POS tags**, **TextBlob sentiment**, **VADER polarity**, and **TF-IDF vectors**—to improve the model's ability to distinguish between *neutral* and *happiness* emotions, a key weakness in Iteration 1.

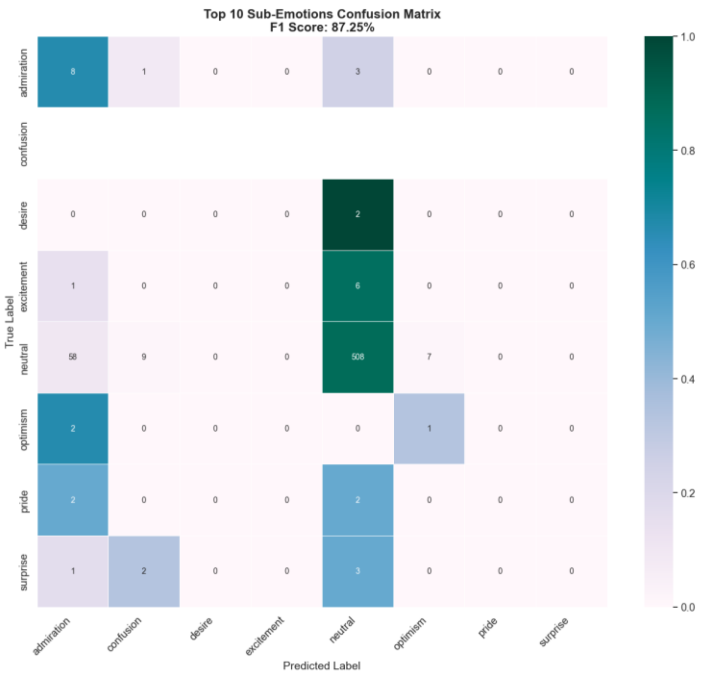

Figure 6: Sub-emotion Confusion Matrix – Iteration 2

While the sub-emotion prediction remained strong (**F1**: **87.3%**), see ***Figure 6*** above, overall improvements were modest. Emotion classification reached an F1 score of **52.7%**, with high accuracy for **happiness (85%)**, but persistent confusion between **neutral**, **disgust**, and **surprise**. **Neutral** continued to dominate in sub-emotion predictions (70% recall), but **nuanced labels** like *curiosity, desire,* and *pride* remained hard to detect. **Intensity performance** saw a slight uptick (F1: **42.6%**), with **neutral intensity** reaching 76% accuracy, shown in ***Figure 7***. However, **intense expressions remained undetected** (0% recall), and **moderate levels** were often mislabelled as mild or neutral.

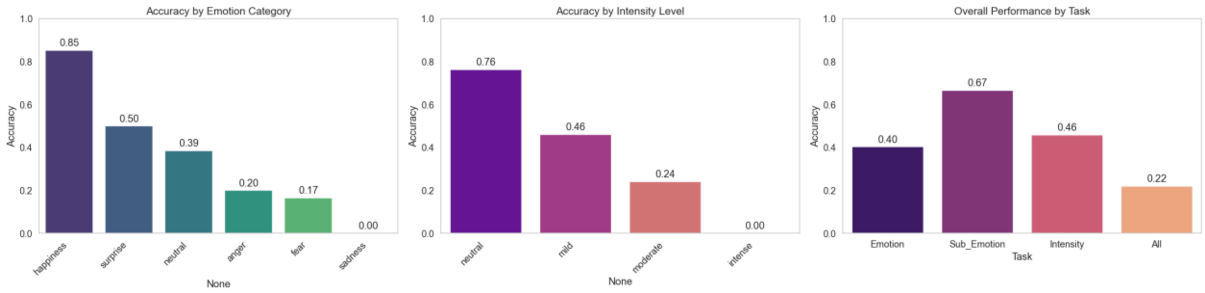

Figure 7: Accuracy - Iteration 2

Misclassification patterns largely mirrored Iteration 1, with **neutral utterances**—such as *"I forgive you"*—often misclassified as **emotional** (e.g., *happiness*), despite correct sub-emotion and intensity predictions, showing that the model still struggles with **emotional subtlety**, especially in **short or unclear** texts.

Figure 8 demonstrates that the integration of linguistic features led to a **28% reduction** in the most frequent error—misclassifying *neutral* as *happiness*—which dropped from 275 to **197 instances**. However, the **average confidence** in these incorrect predictions remained nearly unchanged at **0.539**, indicating the model remains moderately uncertain in ambiguous cases.

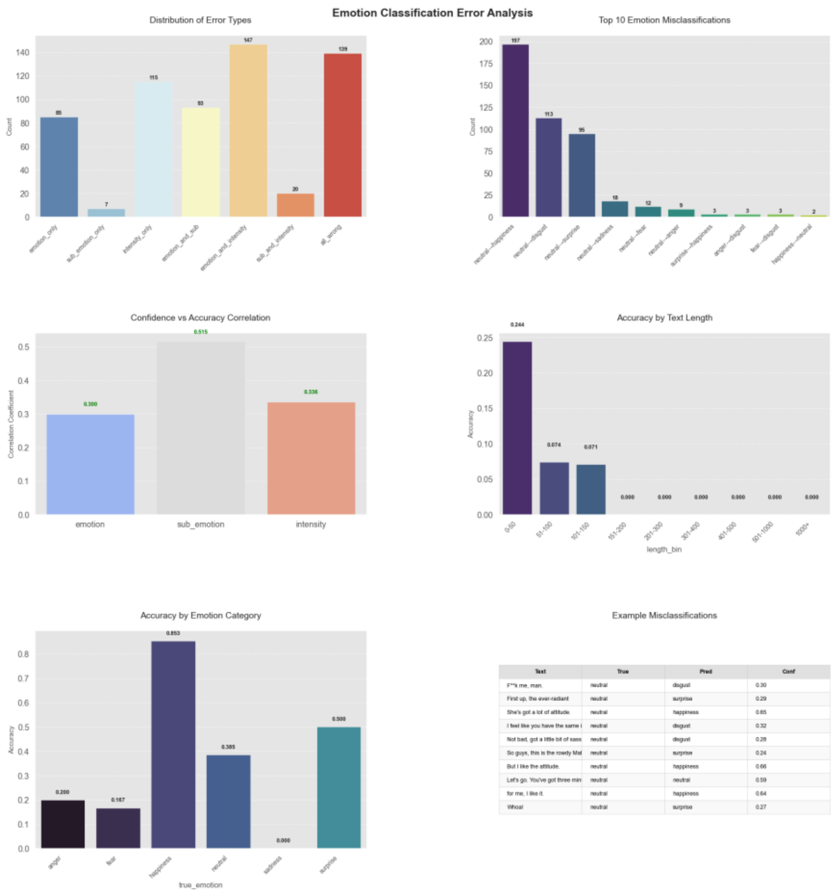

Figure 8: Error Analysis - Iteration 2

Despite these improvements, the **overall accuracy *(Figure 7)*** stayed consistent at **0.22**, suggesting that while one error type improved, others may have emerged. The **neutral class** remained the most error-prone, responsible for **444 out of 606 total errors (73.3%)**. Notably, **sub-emotion accuracy dropped slightly** (from **0.727** to **0.667**), and **intensity accuracy remained stable** (**0.458** vs **0.463**), highlighting the challenge of balancing improvements across hierarchical levels. **Performance by text length** remained unchanged, and **confidence-to-accuracy correlations** were strongest for sub-emotion (**0.515**), followed by intensity (**0.336**) and emotion (**0.300**).

These results suggest that while **feature extraction** helped mitigate dominant error patterns, further progress required better handling of **nuanced** and **long-form expressions**, which can be addressed through **hierarchical attention**, **semantic augmentation**, and **contrastive learning**.

Iteration 3

In the third iteration, we extended the feature pipeline from Iteration 2 by adding **EmoLex (NRC Emotion Lexicon)** features to better capture emotional nuances in text. This aimed to address the ongoing confusion between **neutral** and **happiness** emotions and improve upon the previous iteration's **22% accuracy** plateau.

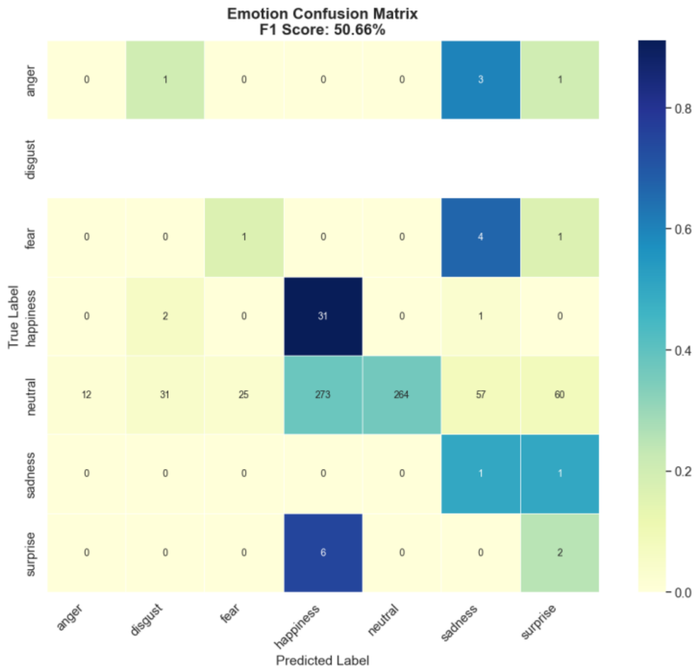

Figure 9: Emotion Confusion Matrix – Iteration 3

This change led to a **slight drop** in **emotion classification performance**, with the **F1 score decreasing from 52.7% to 50.7%** and **accuracy falling from 0.40 to 0.38**. While **happiness** accuracy improved from **85%** to **91%**, **neutral** continued to be the most misclassified, with **273 instances** wrongly labelled as **happiness**—a **rise from 197** in Iteration 2. Misclassifications also spread more toward **sadness (57)** and **surprise (60)**, as depicted in ***Figure 9***, showing that neutral utterances continued to be interpreted as overtly emotional.

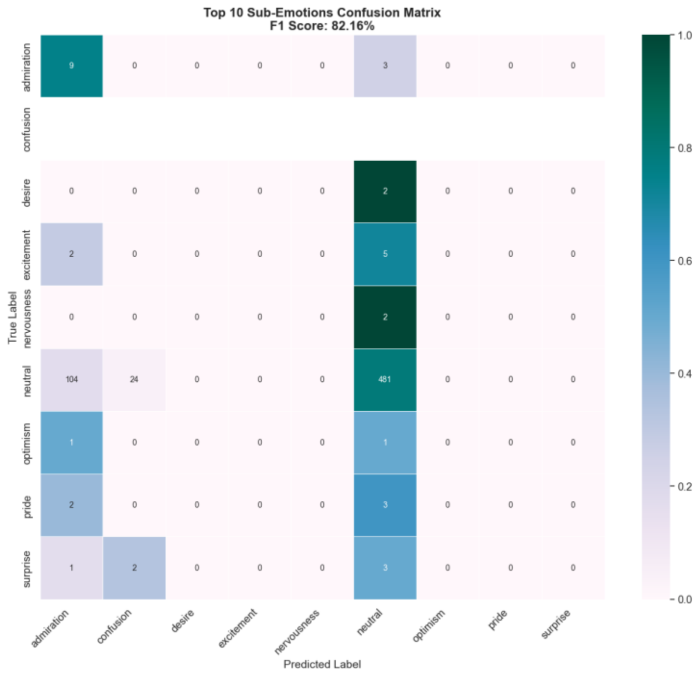

Figure 10: Sub-emotion Confusion Matrix – Iteration 3

Sub-emotion performance also declined, with the **F1 score falling from 87.3% to 73.4%** and **neutral recall** dropping slightly from **70% to 66%**. Nuanced sub-emotions like **optimism**, **desire**, and **pride** remained undetected, shown in ***Figure 10*** above. However, the model showed some increased sensitivity to **admiration**, which appeared frequently as a predicted label for emotional misclassifications. **Intensity classification** remained **stable** (**F1: 42.6% → 42.7%**), with the best performance again in the **neutral** class (**71% accuracy**, down from **76%**), followed by **mild (51%)**, and **moderate (24%)**. **Intense expressions** were still not detected (**0% recall**).

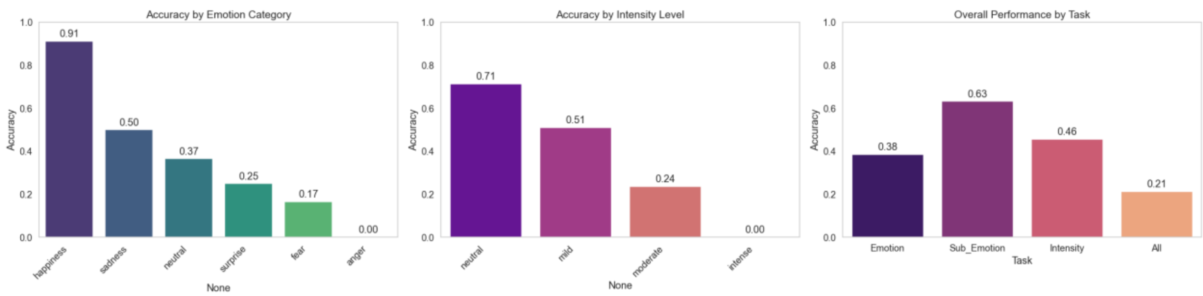

Figure 11: Accuracy - Iteration 3

Figure 11 demonstrates that overall performance across tasks showed little change, with **total accuracy slipping from 0.22 to 0.21**. **Neutral-to-happiness misclassifications** shown below in ***Figure 12*** accounted for nearly **75% (458 of 611)** of all errors, reinforcing neutral as the most unstable class. The model also became **more confident in its mistakes**, with average confidence in misclassifications rising to **0.585** (from **0.539**), indicating a shift toward **overconfident but wrong predictions**. This was mirrored in the sharp drop in **confidence-to-accuracy correlation** for emotion predictions (**0.082**, down from **0.300**), while **sub-emotion (0.510)** and **intensity (0.320)** correlations stayed stable.

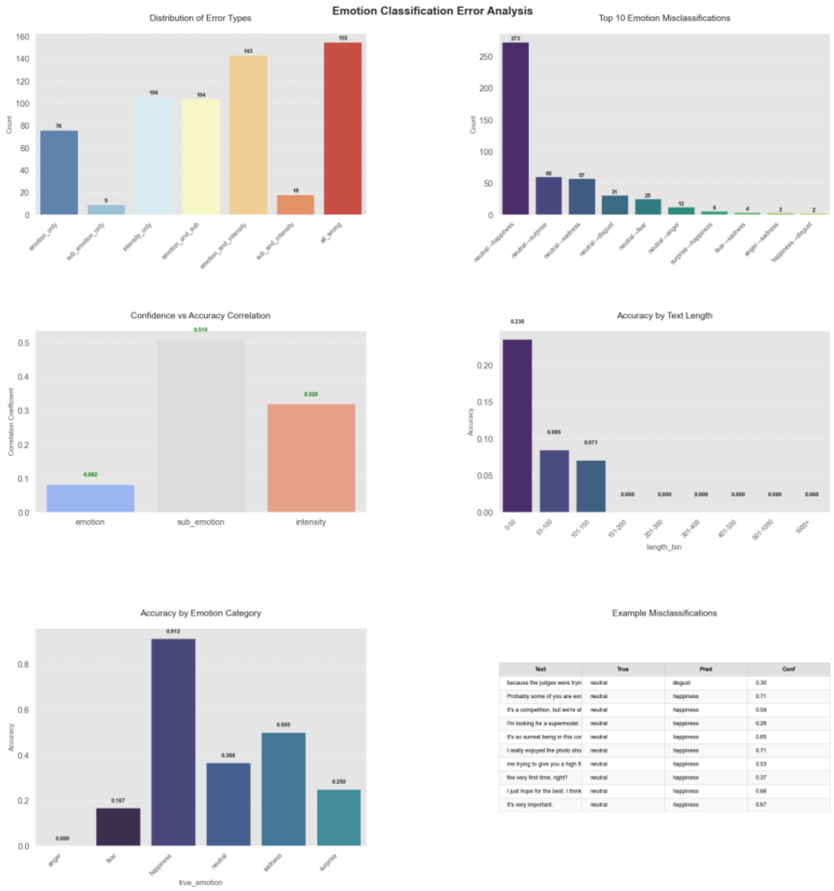

Figure 12: Error Analysis - Iteration 3

Misclassifications were often triggered by **positive phrasing in neutral statements**, such as *"because I'm a very hardworking girl"* or *"this is a dream come true"*— which were frequently mislabelled as **happiness** with sub-emotions like **admiration**. Neutral comments about effort or context showed similar errors, revealing an **overreliance on sentiment cues**. Accuracy remained limited to **very short inputs (<50 characters)**, reinforcing earlier trends. Overall, the model became more decisive but remained misaligned with emotional nuance, underscoring the limits of **lexicon-based features**.

Iteration 4

In the fourth iteration, we returned to the **simpler baseline** setup from Iteration 1, removing all **handcrafted features** to reduce complexity. Instead of adding new input features, we introduced a **post-processing method** that mapped predicted **sub-emotions** to their corresponding **parent emotions**. This **hierarchical correction** aimed to leverage the model's **stronger sub-emotion predictions** to refine emotion outputs, addressing the persistent **neutral-to-happiness misclassification** issue and improving overall **emotion classification** without retraining the model.

The introduction of **sub-emotion–to–emotion mapping** led to a **substantial performance boost**. **Emotion accuracy jumped from 0.38 to 0.734**, and the **weighted F1 score reached 0.801**—the highest across all iterations. The number of **neutral predictions increased to 597**, while **happiness dropped to 144**, directly addressing the overprediction seen in earlier rounds. The predicted emotion distribution aligned more closely with the true labels, especially for the dominant **neutral class (74.2% predicted vs. 93.2% true)**, which can be seen in ***Figure 13*** below.

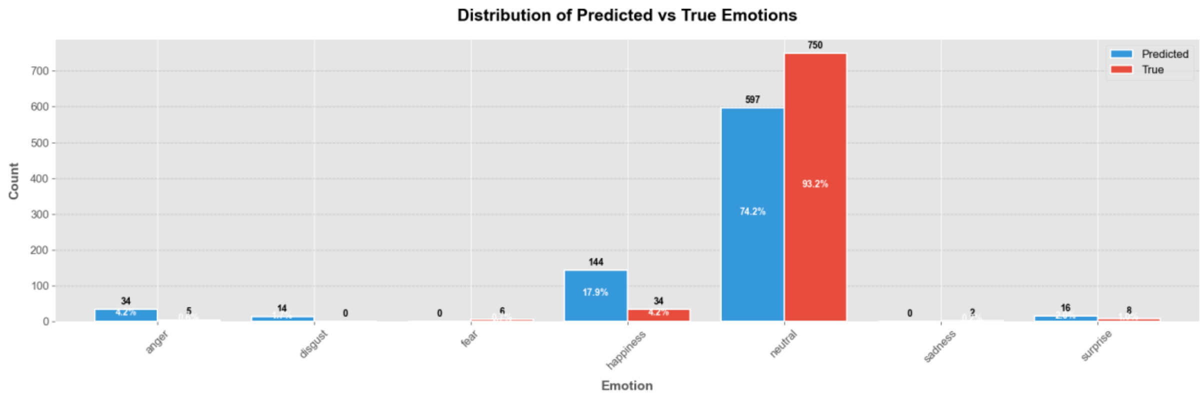

Figure 13: True vs. Predicted Label Distribution - Iteration 4

While classification of **neutral** emotions improved significantly—achieving **precision of 0.96** and an **F1 score of 0.85**— gains came at the cost of sensitivity to **minority emotions**. **Happiness recall dropped to 41%**, and classes like **fear, sadness, and disgust** showed near-zero recall, which is evident in ***Figure 14***. The model also achieved stronger balance between **precision (0.8995)** and **recall (0.7342)**, contributing to a **lower cross-entropy loss of 1.4313**. These results demonstrate that **hierarchical post-processing**, even without additional features or retraining, can **meaningfully improve top-level emotion predictions**, especially in the face of class imbalance.

F1 Score: 0.8011
Precision: 0.8995
Recall: 0.7342
Accuracy: 0.7342
Loss: 1.4313

	precision	recall	f1-score	support
anger	0.09	0.60	0.15	5
disgust	0.00	0.00	0.00	0
fear	0.00	0.00	0.00	6
happiness	0.10	0.41	0.16	34
neutral	0.96	0.76	0.85	750
sadness	0.00	0.00	0.00	2
surprise	0.06	0.12	0.08	8
accuracy			0.73	805
macro avg	0.17	0.27	0.18	805
weighted avg	0.90	0.73	0.80	805

Figure 14: Classification Report - Iteration 4

However, ***Figure 15*** shows that **neutral-to-happiness remained the top error** with **275 instances**, making up nearly **45.5% of all mistakes**. Many involved **short, ambiguous statements** (e.g., “Yes!”, “Good job”), often paired with **sub-emotions like admiration**, leading to consistent overprediction of happiness. These errors were made with **moderate confidence (avg: 0.552)**, slightly **lower than in Iteration 3 (0.585)**, indicating the model was still **uncertain** in borderline cases. Confidence-to-accuracy correlation remained strongest for **sub-emotion (0.508)**, while **emotion (0.306)** and **intensity (0.293)** correlations were moderate.

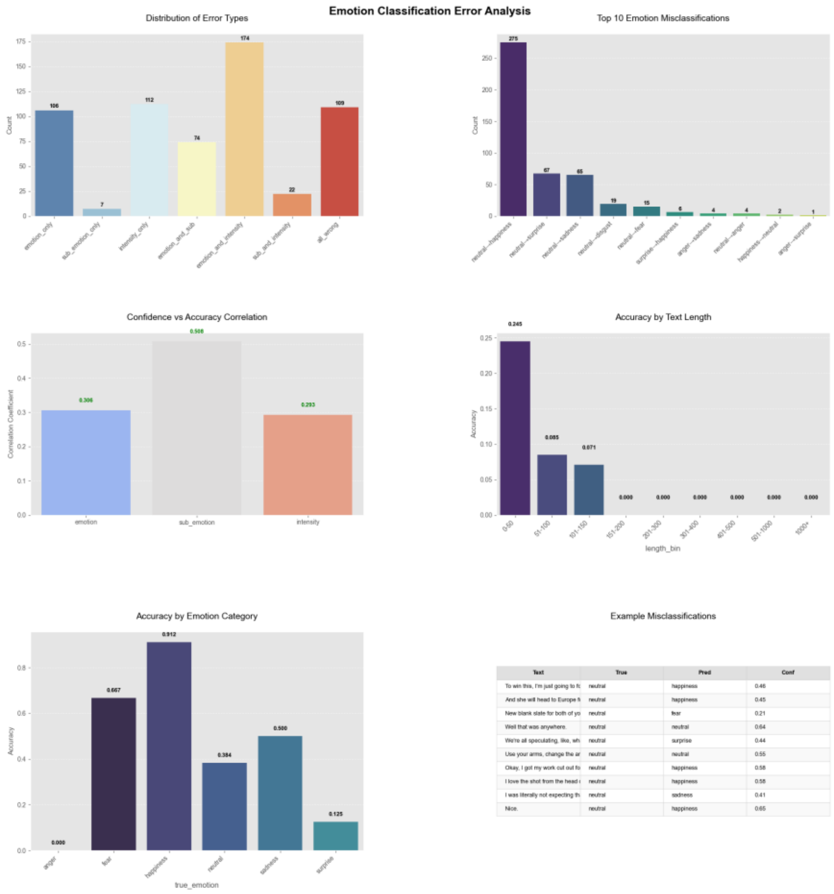

Figure 15: Error Analysis - Iteration 4

Despite improvements in top-line metrics, the **macro F1 score remained low (0.18)**, and errors remained concentrated in the neutral class. Performance was still limited to **very short inputs (<50 characters)**, reinforcing earlier findings that emotional nuance and minority class detection remain unresolved challenges.

Iteration 5

In Iteration 5, we aimed to improve **model robustness** and **minority class performance** through two key changes: applying **class-targeted text augmentation** to increase representation of underrepresented sub-emotions, and using the **raw university-provided test set** instead of the filtered version. The raw set offered a broader and more balanced distribution of emotion classes, enabling more **rigorous and realistic evaluation** of model performance.

To build on the best results from Iteration 1, we retained the **DeBERTa-v3-small backbone** and reintroduced **TF-IDF and EmoLex features**, this time alongside the augmented dataset. We simplified intensity to **three levels** (mild, moderate, strong) and trained a **custom BERT classifier** from scratch with the updated data and features—aiming to address class imbalance and overfitting while improving generalization.

F1 Score: 0.7827
Precision: 0.7883
Recall: 0.7839
Accuracy: 0.7839
Loss: 1.3816

	precision	recall	f1-score	support
anger	0.39	0.75	0.51	16
disgust	0.65	0.53	0.59	32
fear	0.79	0.53	0.63	36
happiness	0.83	0.88	0.85	370
neutral	0.79	0.77	0.78	253
sadness	0.69	0.55	0.61	33
surprise	0.76	0.69	0.73	65
accuracy			0.78	805
macro avg	0.70	0.67	0.67	805
weighted avg	0.79	0.78	0.78	805

Figure 16: Classification Report - Iteration 5

Figure 16 proves that the model achieved an **emotion classification accuracy of 0.784** and a **weighted F1 score of 0.783**, closely matching the performance peak seen in Iteration 4. However, unlike the previous round—where neutral dominance inflated scores—this result reflects a **more balanced distribution across all emotions**. The **macro F1 score rose sharply to 0.67** (from 0.18), signalling improved performance on minority classes. **Happiness (F1: 0.85)** and **neutral (F1: 0.78)** remained strong, while **surprise (0.73)** and **fear (0.63)** saw marked improvements. Even **anger**, previously undetectable, reached **0.51**, indicating a more equitable treatment of emotional classes. The overall **loss decreased to 1.34**, confirming stronger alignment between predictions and true labels.

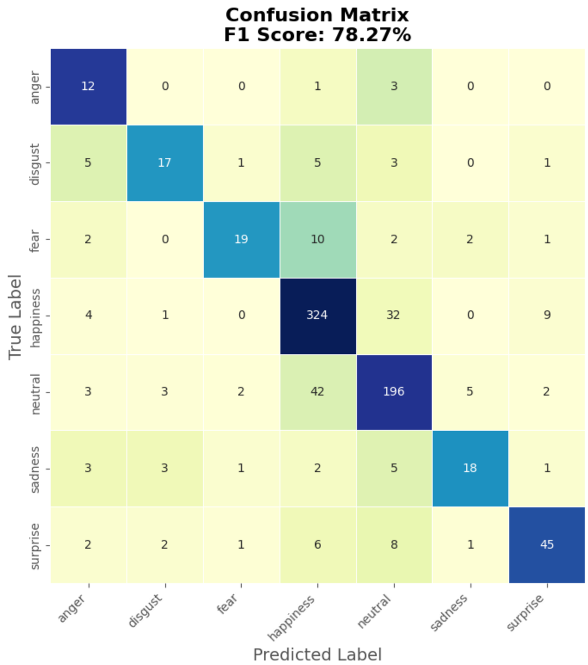

Figure 17: Emotion Confusion Matrix - Iteration 5

Error analysis showed reduced confusion between **neutral and happiness**, and misclassifications were no longer dominated by a single error type. The confusion matrix in ***Figure 17*** above confirmed more **distributed mistakes**, with fewer false positives in short, ambiguous inputs. This shift suggests that **data diversity** and **augmentation strategies** not only improved class balance but also helped the model develop finer distinctions between similar expressions. While neutral still posed challenges, the model was less biased toward it, and accuracy extended beyond the shortest input lengths—marking a meaningful leap in robustness and generalization.

Conclusion

Over five iterations, our approach evolved from simple baseline modelling to sophisticated post-processing and data-driven strategies. While early feature engineering (Iterations 2–3) offered limited gains, the real breakthrough came in **Iteration 4**, where **hierarchical emotion mapping** capitalized on the model's high sub-emotion accuracy to dramatically boost overall emotion classification (**F1: 0.801**).

In **Iteration 5**, we shifted focus to **data augmentation** and reinstated key features (**TF-IDF, EmoLex**), yielding a robust, well-generalized model (**F1: 0.783**) with improved performance across **minority classes** and more balanced predictions.

The journey highlights a key insight: **simpler architectures paired with strategic post-processing or data-centric approaches can outperform more complex feature-heavy pipelines**. By learning from performance patterns and error behaviour at each stage, we systematically refined our model to be not just more accurate, but also fairer and more adaptable across real-world emotional expressions.

Disclosure

ChatGPT (OpenAI, 2024) was used solely to refine language and improve clarity in this report. All interpretations, findings, figures and conclusions remain our own. (OpenAI. (2024). ChatGPT (April 2024 version) [Large language model]. https://openai.com/chatgpt)